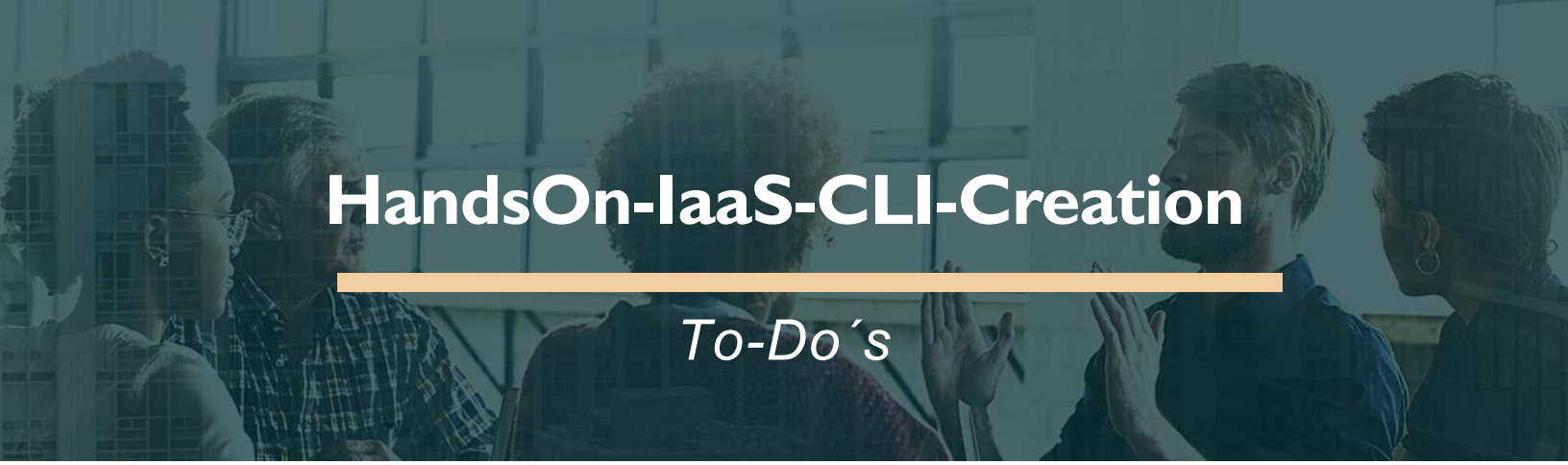


HandsOn-IaaS-CLI-Creation

To-Do´s

Prerequisites

- ☐ **1.** Ensure that you have access to an active Azure Student Subscription. You can get it by calling the following URL and follow the instructions: https://azure.microsoft.com/en-us/free/students/
  IMPORTANT: Use your FHNW account (e.g. Thomas.etteler@students.fhnw.ch)!
- ☐ **2.** Ensure that your browser of choice is set to English as a display language. Otherwise you might get issues when using the Azure Portal.
- ☐ **3.** Ensure that you have a code editor – like Visual Studio code (https://code.visualstudio.com/download) – installed on your machine.

General Settings

- ☐ **1.** Due to performance reasons, please create all Azure resources within the Region Switzerland North.
- ☐ **2.** All resources may require a three-digit identifier, e.g. TET for **T**ohmas **ET**eller. Please think of your identifier before you start with creating resources, it will save you time.
  Your personal 3-digit identifier: ___ ___ ___

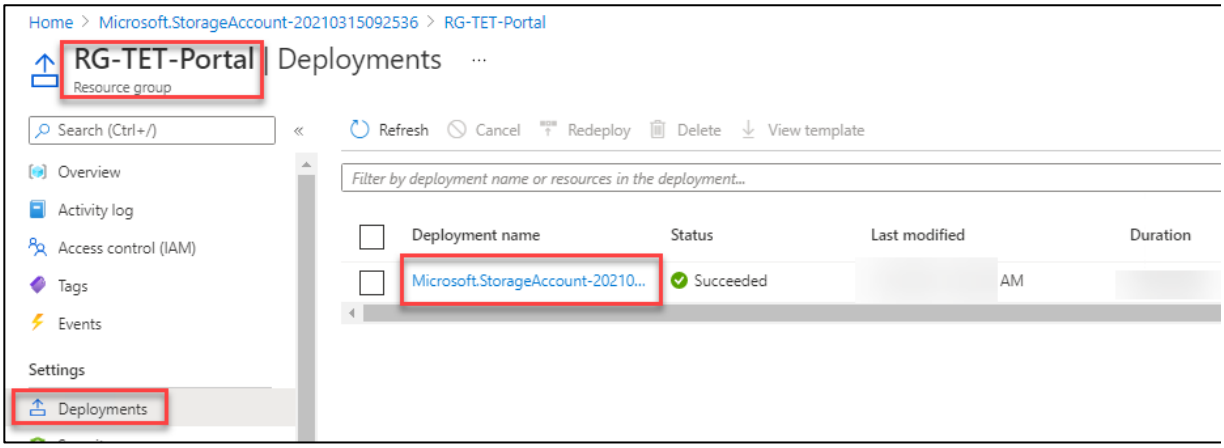
Azure IaaS – Action items

☐ **1. Connect to the Azure Portal - ~1 Min.**

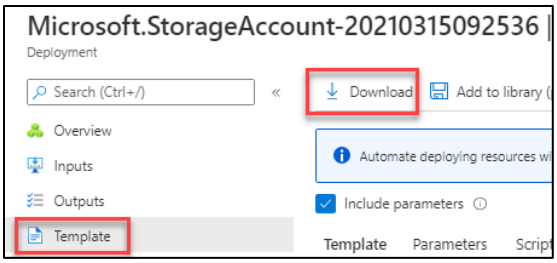
Use your browser of choice and connect to the Azure portal at
https://portal.azure.com/
Authenticate using your FHNW credentials (e.g. sth. Like Thomas.etterler@students.fhnw.ch)

☐ **1. Get the deployment template and modify it - ~5 Min.**

Since we already deployed a storage account, we could leverage this effort to get the provisioning code instead of doing everything ourselves.
Therefore navigate into the resource group we have created in part one, in there go to the Settings-> Deployments. Select the deployment as shown in the screenshot below.



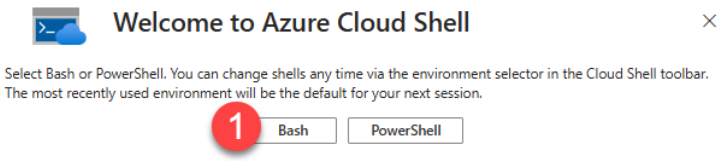
Once you are in the Deployment details switch into the Template section and select "Download" to get the template and the property file. Store it on your device within a location you remember (e.g. C:\Temp\). Unzip the file in the folder.



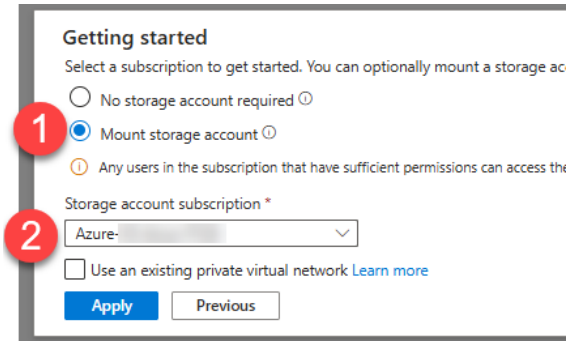
In the unzipped folder you will find two files: parameters.json and template.json. Use your (text-)editor of choice and open both of them and check what is in there. In the parameters file locate the property "storageAccountName" and change the ending from "portal" to "cli" and save the file. This will ensure that your new storage account has a different name.

2. Connect to the Cloud Shell - ~7 Min.

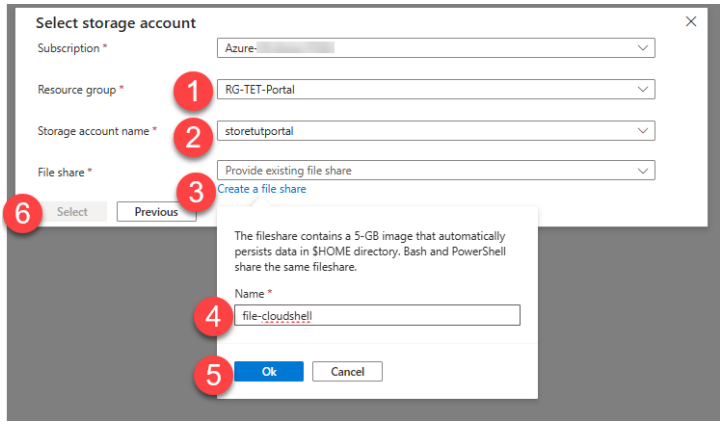
User your browser of choice and connect to the URL:
https://shell.azure.com
and authenticate yourself by using your FHWN Credentials.
Chose “Bash” in the appearing dialog.



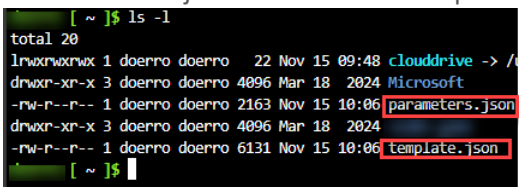
In the next screen, select “Mount storage account” (1) and select the right Subscription in the Dropdown (2).
Take the first option in the next window and click next.



To select the storage account, 1st you have to select the resource group you have created in Hands-On 1, as well as the storage account name (2). The file share needs to be created newly. For this click (3) enter the name “file-cloudshell” (4) and click OK. Finally create the cloud shell by clicking on “Select” (6).



Your cloud shell is ready once the prompt is blinking. To upload the infrastructure-as-code (IaC) files, you can use the “Manage files” option the Cloud Shell Ribbon Bar (3rd Option) and select “Upload”. Navigate to the location on your device where you have stored the adjusted JSON files and upload both of them (template.json and parameters.json). Now you should be able to see these files in your Cloud Shell. To ensure this type the command “`ls -l`” which should give you a similar overview like in the screenshot aside.



3. Deploy the resource using the CLI - ~5 Min.

Now that all requirements are in place, we can start with the command line based deployment. But first of all, we need a new resource group to ensure that we can differentiate between the deployments. You will achieve this, by entering the following command into the Cloud Shell, but make sure you enter your 3-digits first!

```
az group create --name RG-<your-3-digits>-CLI --location switzerlandnorth
```

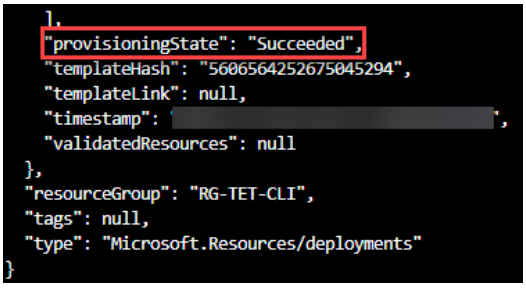
After this task completed successfully (you should see a parameter in the output called "provisioningState" that has the value "Succeeded") you can deploy the storage account resource into it. This is done, by running the following one command within the Cloud Shell.

ATTENTION: It is one line and make sure to replace your 3-digits

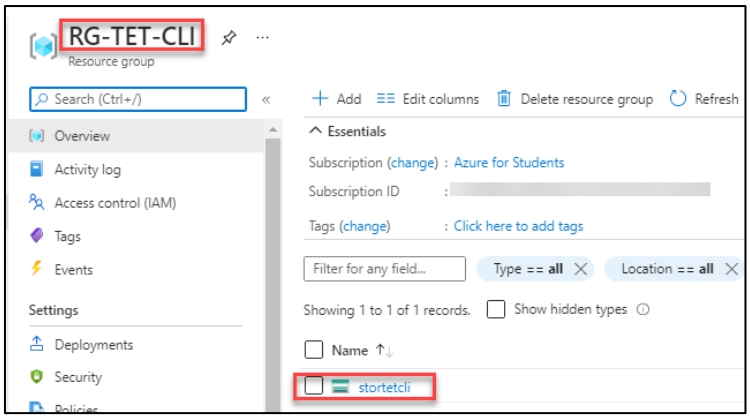
```
az deployment group create --resource-group RG-<your-3-digits>-CLI
--template-file template.json --parameters parameters.json
```

If you would like to know more about the commands, please check out the following web sites:
https://docs.microsoft.com/en-us/azure/azure-resource-manager/templates/deploy-cli
https://docs.microsoft.com/en-us/cli/azure/deployment?view=azure-cli-latest#az_deployment_create

When your task has finished, your command line output should look like this:



Finally we should validate that the resource is visible in the Azure Portal as well. Therefore check your resource groups listing in the Azure Portal. If it´s not visible in there, make sure to hit the "Refresh" Button in the top navigation.



That´s it

You are done with the ***Infrastructure-as-code*** deployment of an Azure storage account.

4. Advanced – Do it with Azure Bicep

If there is time available, you can use the Azure Bicep approach:
https://github.com/mattias-fjellstrom/azure-bicep-upload-data-to-storage
You can use the previously created Azure Cloud Shell to run the Bicep commands.